



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2022 - 2023)

JEE (Main)

FULL TEST

23-01-2023

JEE(Main + Advanced) : ENTHUSIAST COURSE (SCORE-I)

ANSWER KEY

PART-1 : PHYSICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A	C	C	C	A	C	D	A	D	A
	Q.	11	12	13	14	15	16	17	18	19	20
	A.	A	D	C	A	C	A	C	C	C	A
SECTION-II	Q.	1	2	3	4	5	6	7	8	9	10
	A.	6174	364	1	125	4	64	4	10	981	347

PART-2 : CHEMISTRY

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A	D	D	A	D	C	B	A	C	B
	Q.	11	12	13	14	15	16	17	18	19	20
	A.	C	A	C	D	B	D	D	B	D	B
SECTION-II	Q.	1	2	3	4	5	6	7	8	9	10
	A.	5	13	505	4	5	4	4	6	0	19

PART-3 : MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A	D	D	D	B	C	B	D	B	A
	Q.	11	12	13	14	15	16	17	18	19	20
	A.	B	C	C	C	D	C	A	A	B	B
SECTION-II	Q.	1	2	3	4	5	6	7	8	9	10
	A.	5	5	8	4	19	32	2	303	1	2

HINT – SHEET

PART-1 : PHYSICS

SECTION-I

1. Ans (A)

Change in momentum along y-direction will be cancelled out

∴ Change in momentum along x-direction,

$$\Delta p = 2mv \cos \theta$$

$$\text{Pressure} = \frac{\text{Force}}{\text{Area}} = \frac{\frac{\Delta p}{\Delta t}}{A}$$

$$= \frac{\Delta p}{A \cdot \Delta t} = \frac{\Delta p}{A \cdot \frac{L}{v \cos \theta}} = \frac{\Delta p v \cos \theta}{A \cdot L}$$

$$\text{Pressure } p' = \frac{2mv \cos \theta \cdot v \cos \theta N}{V},$$

$$\therefore \left(n = \frac{N}{V} \right), (V = \text{Area} \times L = A \times L)$$

$$p' = 2mnv^2 \cos^2 \theta$$

2. Ans (C)

$$T = 2\pi \sqrt{\frac{l}{g}}$$

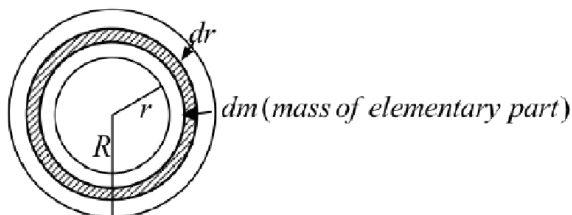
Since, lift accelerated upward, then

$$T' = 2\pi \sqrt{\frac{l}{g+g'}} = 2\pi \sqrt{\frac{l}{g+\frac{g}{4}}}$$

$$= 2\pi \sqrt{\frac{l}{5g}} \times 4 = 2\pi \sqrt{\frac{l}{g}} \times \frac{2}{\sqrt{5}} = \frac{2T}{\sqrt{5}}$$

3. Ans (C)

$$\text{Pressure } dp = \frac{dm \cdot g}{A} = \frac{dm \cdot g}{4\pi r^2} = \frac{\rho \cdot 4\pi r^2 dr GM \frac{r}{R^3}}{4\pi r^2}$$



$$dp = \frac{\rho \cdot 4\pi r^2 dr G \cdot \rho \cdot \frac{4\pi}{3} R^3 \frac{r}{R^3}}{4\pi r^2} \Rightarrow dp = \frac{4\pi}{3} \rho^2 G r dr$$

$$\int dp = \int_r^R \frac{4\pi}{3} \rho^2 G r dr \Rightarrow p = \frac{4\pi}{3} \rho^2 G \left(\frac{r^2}{2} \right)_r^R$$

$$\Rightarrow p = \frac{4\pi}{3} \rho^2 G \left(\frac{R^2}{2} - \frac{r^2}{2} \right)$$

$$\Rightarrow p = \frac{4\pi}{3} \rho^2 G \left(\frac{R^2 - r^2}{2} \right)$$

$$\Rightarrow p = \frac{2\pi}{3} \rho^2 G (R^2 - r^2)$$

4. Ans (C)

$$\frac{mdv}{dt} = mg + kv^2 \Rightarrow \frac{dv}{dt} = g + \frac{k}{m} v^2$$

$$\Rightarrow \frac{dv}{g + \frac{k}{m} v^2} = dt$$

$$\Rightarrow \int \frac{dv}{g + \frac{k}{m} v^2} = \int dt \Rightarrow \int \frac{dv}{\frac{k}{m} \left(\frac{gm}{k} + v^2 \right)} = \int dt$$

$$\Rightarrow \frac{m}{k} \times \frac{1}{\sqrt{\frac{gm}{k}}} \tan^{-1} \frac{v}{\sqrt{\frac{gm}{k}}} = t$$

$$\Rightarrow v = \sqrt{\frac{gm}{k}} \tan \left(\sqrt{\frac{kg}{m}} \cdot t \right)$$

5. Ans (A)

Magnetic flux through the loop $\phi = NBA$

$$\text{induced e.m.f., } \varepsilon = - \frac{d\phi}{dt}$$

and induced current

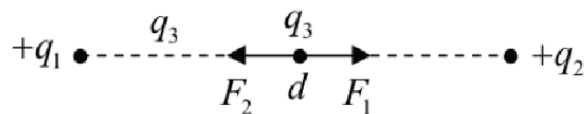
$$i = - \frac{1}{R} \frac{d\phi}{dt} = \frac{dQ}{dt} \Rightarrow - \frac{1}{R} d\phi = dQ$$

$$\text{Thus, } \frac{1}{20} \times (40 \times B \times 0.01) = 2 \times 10^{-5}$$

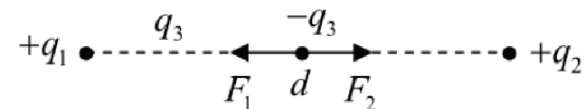
$$\Rightarrow B = 1 \times 10^{-3} \text{ T}$$

6. Ans (C)

If q_3 is positive,



If q_3 is negative,



In both case there is possibility that charge q_3 may be in equilibrium.

7. Ans (D)

Electrostatic potential energy

$$U \propto \frac{1}{r^3} \Rightarrow \frac{U_2}{U_1} = \frac{r_1^3}{r_2^3} = \frac{1}{8}$$

8. Ans (A)

When point P is inside due to motional emf, potential PQ is positive. When point Q is inside potential QP is positive or potential PQ is negative.

9. Ans (D)

Displacement current density

$$J_d = \epsilon_0 \frac{dE}{dt} = \frac{\epsilon_0}{d} \frac{\partial V(t)}{\partial t} = \frac{\epsilon_0 \omega V_0 \sin \omega t}{d}$$

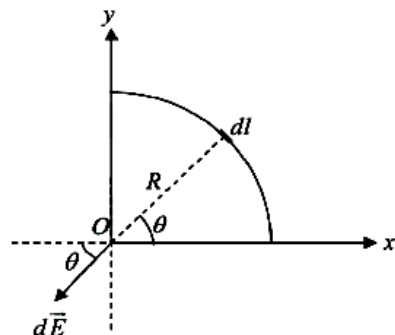
10. Ans (A)

$$\vec{B} = \frac{\vec{k} \times \vec{E}}{\omega} = \frac{k\hat{z} \times E_0 \sin(kz - \omega t)\hat{j}}{\omega}$$

$$= - \frac{kE_0}{\omega} \sin(kz - \omega t)\hat{i}$$

$$= - \frac{E_0}{c} \sin(kz - \omega t)\hat{i}$$

11. Ans (A)



$$\vec{E} = -E_x \hat{x} - E_y \hat{y}$$

$$\text{where } E_x = \int_{\text{line}} dE \cos \theta, \quad E_y = \int_{\text{line}} dE \sin \theta$$

$$dE = \frac{1}{4\pi\epsilon_0} \frac{\lambda dl}{R^2}$$

$$E_x = \int_{\text{line}} \frac{1}{4\pi\epsilon_0} \frac{\lambda dl}{R^2} \cos \theta = \frac{\lambda}{4\pi\epsilon_0} \int_0^{\pi/2} \cos \theta \frac{R d\theta}{R^2}$$

$$E_x = \frac{\lambda}{4\pi\epsilon_0 R} [\sin \theta]_0^{\pi/2} = \frac{\lambda}{4\pi\epsilon_0 R}$$

$$\text{similarly } E_y = \int_{\text{line}} \frac{1}{4\pi\epsilon_0} \frac{\lambda dl}{R^2} \sin \theta = \int_0^{\pi/2} \sin \theta \frac{R d\theta}{R^2}$$

$$\Rightarrow E_y = \frac{\lambda}{4\pi\epsilon_0 R} [-\cos \theta]_0^{\pi/2} = \frac{\lambda}{4\pi\epsilon_0 R}$$

$$\text{Thus } \vec{E} = -E_x \hat{x} - E_y \hat{y} = \frac{\lambda}{4\pi\epsilon_0 R} (-\hat{x} - \hat{y})$$

12. Ans (D)

$$E_n = \frac{q^2 B^2 r_n^2}{2m}$$

$$\because mv_n r_n = n\hbar \text{ and } r_n = \frac{mv_n}{qB}$$

$$\Rightarrow r_n = \frac{m}{qB} \frac{n\hbar}{mr_n} \Rightarrow r_n^2 = \frac{n\hbar}{qB}$$

$$\Rightarrow E_n = \frac{q^2 B^2 r_n^2}{2m} = \frac{q^2 B^2}{2m} \times \frac{n\hbar}{qB} = n \left(\frac{qB\hbar}{4\pi m} \right)$$

13. Ans (C)

$$E_x = E_0 \sin(\omega t - kz), \quad E_y = E_0 \left(\omega t - kz - \frac{\pi}{4} \right)$$

Thus resultant is elliptically polarized wave.

$$\text{At } z = 0, \quad E_x = E_0 \sin(\omega t), \quad E_y = E_0 \sin\left(\omega t - \frac{\pi}{4}\right)$$

$$\text{When } \omega t = 0, \quad E_x = 0, \quad E_y = -\frac{E_0}{\sqrt{2}} \text{ and when}$$

$$\omega t = \frac{\pi}{4}, \quad E_x = \frac{E_0}{\sqrt{2}}, \quad E_y = 0$$

14. Ans (A)

Same intensity since $I_{\min} = 0$

S_1, S_2 must not have equal phase but same phase.

$$\text{as } \phi_1 = \pi/2, \quad \phi_2 = 7\pi/3$$

15. Ans (C)

$$x|_{t=0} = A \cos \theta = 0 \Rightarrow \cos \alpha = 0 \Rightarrow \alpha = \frac{(2n+1)\pi}{2}$$

$$\frac{dx}{dt} \Big|_{t=0} = A e^{-\gamma t} (-\gamma) \cos(\omega t + \alpha) - A e^{-\gamma t} \sin(\omega t + \alpha) \omega$$

$$= A e^{-\gamma t} (-\gamma \cos(\omega t + \alpha) - \sin(\omega t + \alpha) \omega)$$

$$\frac{dx}{dt} \Big|_{t=0} = v \Rightarrow A (-\gamma \cos(\alpha) + \omega \sin(\alpha)) = v$$

$$\Rightarrow A \omega = v \Rightarrow A = \frac{v}{\omega}$$

16. Ans (A)

$$f(x, t) = A \exp\left(\frac{2abxt - a^2 x^2 - b^2 t^2}{c^2}\right) = A \exp\left[\frac{-(ax - bt)^2}{c^2}\right]$$

Phase factor is constant

$$\frac{-(ax - bt)^2}{c^2} = \text{const} \Rightarrow -(ax - bt)^2 = \text{const} \times c^2$$

Taking differentiation, we get

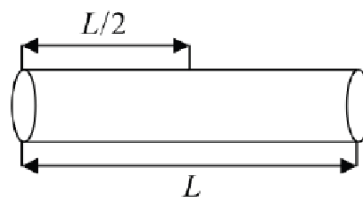
$$(ax - bt)(adx - bdt) = 0 \Rightarrow adx - bdt = 0 \Rightarrow dx/dt = b/a$$

Velocity of the pulse is b/a

17. Ans (C)

The fundamental frequency in organ pipe with

$$\text{both end open is } f = \frac{v}{2L}$$



With additional rate at $\frac{L}{2}$, the fundamental frequency becomes

$$f' = \frac{v}{2L'} = \frac{v}{\frac{2L}{2}} = \frac{v}{L} = \frac{340 \text{ m/sec}}{29 \times 10^{-2} \text{ m}} = 1172 \text{ Hz}$$

18. Ans (C)

$$V_1 = \left(\frac{f}{f - u} \right)^2 V_0 \text{ and } \frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$

For, $u > f$, $v_i > V_0$ and u decreases than v increases.

$\therefore V_1 > V_0$ and image moves away from the lens.

19. Ans (C)

$$C = (\bar{A}\bar{B}) (\bar{A}\bar{B}) \\ = \bar{A}\bar{B} + \bar{A}\bar{B} = A \text{ XOR } B$$

20. Ans (A)

AD is isochoric DC and BA are isothermal and CB is adiabatic.

PART-1 : PHYSICS

SECTION-II

1. Ans (6174)

$$I = \frac{P}{A} = \frac{3.8 \times 10^{26}}{4\pi \times (7 \times 10^{10})^2} \text{ W/cm}^2 = 6174 \text{ W/cm}^2$$

2. Ans (364)

$$I = \frac{V}{R} e^{-t/RC} \\ = \frac{6}{10 \times 10^3} e^{-5/10 \times 10^3 \times 1000 \times 10^{-6}} \\ = \frac{6}{10^4} e^{-5/10} \\ = \frac{6}{\sqrt{e} \times 10^4} = \frac{6}{1.65 \times 10^4} = 364 \mu\text{A}$$

3. Ans (1)

$$E_{OT} = \left(\frac{2n_1}{n_1 + n_2} \right) E_{OI} \\ \Rightarrow \frac{E_{OT}}{E_{OI}} = \left(\frac{2\sqrt{\epsilon_{r1}}}{\sqrt{\epsilon_{r1}} + \sqrt{\epsilon_{r2}}} \right) \\ = \left(\frac{2\sqrt{1.69}}{\sqrt{1.69} + \sqrt{2.25}} \right) = 0.93$$

4. Ans (125)

The condition for constructive interference is

$$2\mu t = \left(m + \frac{1}{2} \right) \lambda, \text{ where } m = 0, 1, 2 \\ \therefore \text{ for } m = 0 \\ \mu = \frac{\lambda/2}{2t} = \frac{\lambda}{4t} = \frac{0.5 \times 10^{-6} \text{ m}}{4 \times 0.1 \times 10^{-6} \text{ m}} = \frac{5}{4} = 1.25$$

5. Ans (4)

Width of central bright fringe.

$$= \frac{2\lambda D}{d} = \frac{2 \times 500 \times 10^{-9} \times 80 \times 10^{-2}}{0.20 \times 10^{-3}} \\ = 4 \times 10^{-3} \text{ m} = 4 \text{ mm.}$$

6. Ans (64)

$$(n_1 - 1)t - (n_2 - 1)t = 7\lambda \\ (n_1 - n_2) = \frac{7\lambda}{t} = \frac{7 \times 550 \times 10^{-9}}{6 \times 10^{-6}} = 0.64$$

7. Ans (4)

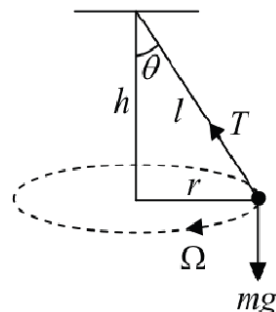
$$u = -30 \text{ cm} \\ f = -20 \text{ cm} \\ \frac{1}{v} + \frac{1}{u} = \frac{1}{f} \Rightarrow \frac{1}{v} = \frac{1}{f} - \frac{1}{u} = \frac{1}{-20} - \frac{1}{-30} \\ \frac{1}{v} = -\frac{1}{60} \Rightarrow v = -60 \text{ cm} \\ m = \frac{I}{O} = -\frac{v}{u} \\ I = -\frac{(-60)}{(-30)} \times 2 \text{ cm} = -4 \text{ cm}$$

8. Ans (10)

$$T \cos \theta = mg \dots\dots\dots(i)$$

Since the object is executing a circular orbit, radius r , with angular velocity Ω , it experiences a centripetal force $m\Omega^2 r$

$$\therefore T \sin \theta = m\Omega^2 r$$



From (i) and (ii)

$$\tan \theta = \frac{\Omega^2 r}{h} \dots\dots\dots(iii)$$

$$\text{since, } \tan \theta = \frac{r}{h} \dots\dots\dots(iv)$$

$$\therefore \frac{r}{h} = \frac{\Omega^2 r}{h} = \Omega = \sqrt{\frac{g}{h}}$$

Now, $h = l \cos \theta$

$$\Omega = \sqrt{\frac{g}{l \cos \theta}}$$

Given $g = 10 \text{ ms}^{-2}$, $l = 20 \text{ cm} = 20 \times 10^{-2} \text{ m}$

$$\theta = \frac{\pi}{3} = 60^\circ$$

$$\Omega = \sqrt{\frac{10}{20 \times 10^{-2} \times \cos(60^\circ)}} = \sqrt{\frac{10}{20 \times 10^{-2} \times \frac{1}{2}}} = \sqrt{100}$$

$$\therefore 10 \text{ rad/s}$$

9. Ans (981)

$$l = 10\text{m}$$

Mass to be pulled

$$\text{Mass of small elementary } \frac{m}{l} \times dy$$

$$\text{PE of mass} = \frac{m}{l} \times dy \times y \times g$$

So work required in pulling

$$W = - \int_1^l dU = - \int_0^l -\frac{m}{l} y dy \times g$$

$$= \frac{m}{l} \times \frac{l^2}{2} \times g = \frac{2 \times 10 \times 9.81 \times 10}{2} = 981\text{J}$$

10. Ans (347)

Let the final temperature of reservoir be T_f

Change in the entropy of source

$$\Delta S_1 = \int \frac{dQ}{T} = \int_{402}^{T_f} \frac{C_p dT}{T} = C_p \log \frac{T_f}{402}$$

Change in entropy of sink

$$\Delta S_2 = \int \frac{dQ}{T} = \int_{300}^{T_f} \frac{C_p dT}{T} = C_p \log \frac{T_f}{300}$$

Since, process is adiabatic, the net change in the entropy is zero.

$$\Delta S_1 + \Delta S_2 = 0$$

$$C_p \log \frac{T_f}{402} + C_p \log \frac{T_f}{300} = 0$$

$$\log \frac{T_f^2}{402 \times 300} = 0$$

$$T_f = \sqrt{402 \times 300} = 347.27\text{K}$$

PART-2 : CHEMISTRY

SECTION-I

1. Ans (A)

$$-\Delta_r G^\circ = 2.303 RT \log k_{eq}$$

$$\Rightarrow \log k_{eq} = - \frac{\Delta_r H^\circ}{2.303R} \left(\frac{1}{T} \right) + \frac{\Delta_r S^\circ}{2.303R}$$

slope = (+)ve $\Rightarrow \Delta_r H^\circ < 0$ intercept (+)ve, (-)ve or zero

2. Ans (D)

$$\text{Rate} = k[N_3^-][CH_3I]$$

$$\text{if } [N_3^-] \gg [CH_3I]$$

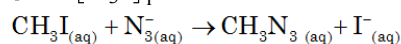
then rate = $k'[CH_3I] \Rightarrow$ 1st order with

$$t_{1/2} = \frac{0.693}{k'} \text{ where } k' = k[N_3^-]$$

so $t_{1/2}$ of CH_3I depends on $[N_3^-]$ taken

Reaction is irreversible so excess I^- doesn't affect reaction rate significantly

$$\text{for } [CH_3I]_i = [N_3^-]_i = a = 0.02\text{M}$$

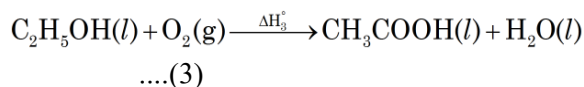
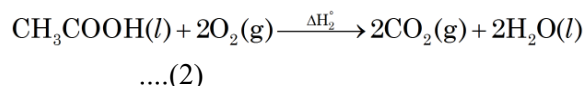
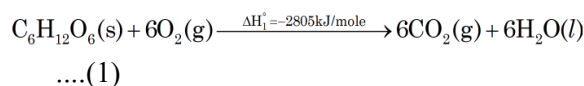


$$\text{At time } t \quad (a-x) \quad (a-x) \quad x \quad x \text{ mol/lit}$$

$$\text{rate} = k(a-x)^2$$

$$\Rightarrow t \Rightarrow t_{1/2} \propto \frac{1}{a}$$

3. Ans (D)



$rk^n (1) - 2 \times [rk^n (2) + rk^n (3)]$ gives the desired reaction for fermentation of glucose

$$\therefore \Delta_r H^\circ = \Delta_r H_1^\circ - 2(\Delta H_2^\circ + \Delta H_3^\circ)$$

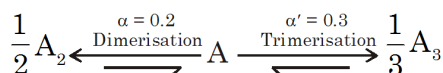
$$= -2805 - 2 \times (-874) - 2(-493)$$

$$= -2805 + 1748 + 986$$

$$= -71 \text{ kJ/mole}$$

4. Ans (A)

In 1kg of solvent :



$$\frac{a\alpha}{2} \quad a - a\alpha - a\alpha' \quad \frac{a\alpha'}{3}$$

$$\frac{1}{2} \times 0.2 \times 0.2 \quad 0.2(1 - 0.2 - 0.3) \quad \frac{1}{3} \times 0.3 \times 0.2$$

total moles of solute after association

$$= 0.2(1 - 0.5) + 0.02 + 0.02$$

$$= 0.1 + 0.04 = 0.14 \text{ moles/kg}$$

$$\Delta T_b = k_b m' = 5 \times 0.14 = 0.7^\circ\text{C}$$

5. **Ans (D)**

Fact

6. **Ans (C)**

Fact

7. **Ans (B)**

$$T = \frac{2\pi r_{n,z}}{V_{n,z}} \propto \frac{n^3}{z^2}$$

$$\therefore \frac{T_{1,2}}{T_{2,z}} = \frac{1^3/2^2}{2^3/z^2} = \frac{9}{32}$$

$$\Rightarrow z = 3$$

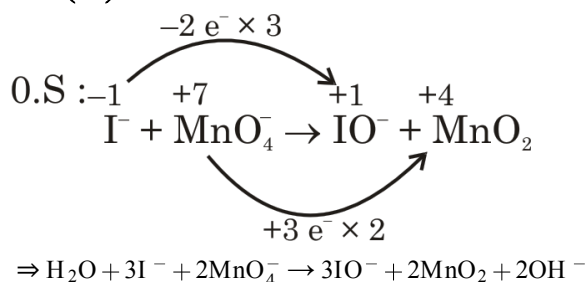
19. **Ans (D)**

Product R is $\text{CH}_3 - \text{CH} = \text{CH} - \text{CHO}$

PART-2 : CHEMISTRY

SECTION-II

1. **Ans (5)**



sum of coefficients of I^- & OH^- = 3 + 2 = 5

2. **Ans (13)**

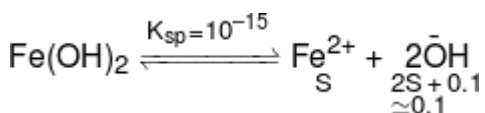
$$E_{\text{OH}|\text{Fe}(\text{OH})_2|\text{Fe}}^\circ = E_{\text{Fe}^{2+}/\text{Fe}}^\circ + \frac{0.06}{2} \log K_{\text{sp}}$$

$$\Rightarrow -0.89 = -0.44 + \frac{0.06}{2} \log K_{\text{sp}}$$

$$\Rightarrow \log K_{\text{sp}} = -15$$

$$\Rightarrow K_{\text{sp}} = 10^{-15}$$

solubility of $\text{Fe}(\text{OH})_2$ in 0.1 M NaOH solution



$$K_{\text{sp}} = 10^{-15} = \text{S}(0.1)^2$$

$$\text{S} = 10^{-13} \text{ mol/lit} = 1 \times 10^{-13} \text{ mol/lit} = x \times 10^{-y}$$

$$\therefore y = 13$$

3. **Ans (505)**

For argon gas : $P_1 V_1 = P_2 V_2$

$$\Rightarrow 1 \times 1 = P_2 \times 5$$

$$\Rightarrow P_2 = 1/5 \text{ atm} = 0.200 \text{ atm}$$

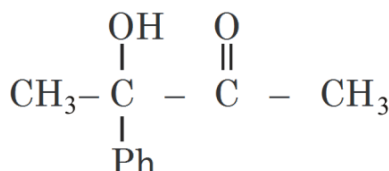
final pressure = $P_{\text{Ar}} + \text{V.p of acetone}$

$$= 0.200 + 0.305 = 0.505 \text{ atm}$$

$$= 505 \text{ milli atm}$$

5. **Ans (5)**

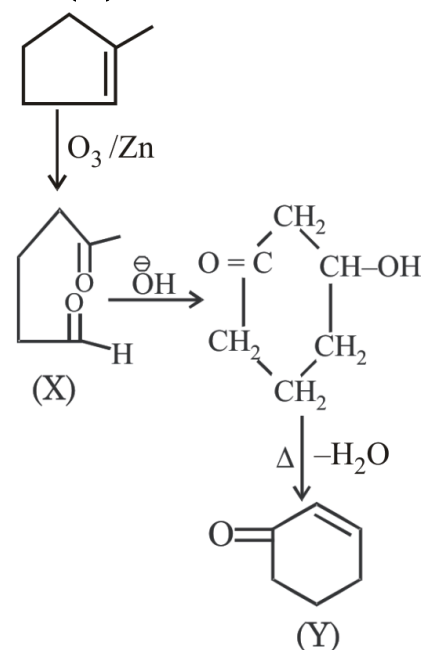
Organic product (Z) is



6. **Ans (4)**

$$x = 3, y = 2, z = 3$$

7. **Ans (4)**



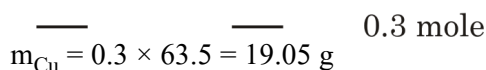
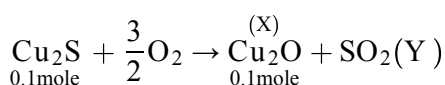
8. **Ans (6)**

Metal is Cr, metal ion : Cr^{+2}

9. **Ans (0)**

Sodium nitroprusside gives colored solution not ppt.

10. **Ans (19)**



PART-3 : MATHEMATICS

SECTION-I

1. Ans (A)

Line is perpendicular to the plane, hence the image will be the same line.

2. Ans (D)

$$\begin{aligned} \int_{-2}^2 \frac{e^x - e^{-x}}{e^x + e^{-x}} dx &= 0 \quad [\text{odd function}] \\ \Rightarrow \int_{-2}^2 \frac{[\cos x] dx}{e^x + e^{-x}} &= \int_{-2}^{-\pi/2} \frac{-1}{e^x + e^{-x}} + 0 + 0 + \int_{\pi/2}^2 \frac{-1 dx}{e^x + e^{-x}} \\ &= 2 \int_2^{\pi/2} \frac{1}{e^x + e^{-x}} = 2 \tan^{-1} e^x \Big|_2^{\pi/2} \\ &= 2 [\tan^{-1} e^{\pi/2} - \tan^{-1} e^2] \end{aligned}$$

3. Ans (D)

$$\begin{aligned} y = mx + 2 \text{ is tangent} &\Rightarrow 2 = \frac{1}{m} \Rightarrow m = \frac{1}{2} \\ \Rightarrow L : x - 2y + 4 &= 0 \quad \dots(i) \\ L_1 : 2x + y - 4 &= 0 \quad \dots(iii) \\ \Rightarrow y^2 + 2y - 8 &= 0 \\ \Rightarrow (y + 4)(y - 2) &= 0 \Rightarrow P(4, -4) \\ \text{Tangent at P is } -4y &= 2(x + 4) \\ \Rightarrow -2y &= x + 4 \quad \dots(ii) \\ Q \text{ is } (1, 2) &\Rightarrow T_Q y = x + 1 \\ \Rightarrow \text{intersection point of L and } T_P &\text{ is } (-4, 0) \\ \text{Also intersection point of L and } T_Q &\text{ is } (2, 3) \\ \Rightarrow \text{Equation of circle is } (x - 2)(x + 4) + y(y - 3) &= 0 \\ \Rightarrow x^2 + y^2 + 2x - 3y - 8 &= 0 \end{aligned}$$

4. Ans (D)

$$\begin{aligned} \alpha = 5, \beta = 3 \\ \text{Equation of line is } 10x + 6y + 22 &= 0 \\ \text{or} \\ 5x + 3y + 11 &= 0 \\ r &= \frac{11}{\sqrt{34}} \\ (1 - i)\bar{z} + (1 + i)z + 4 &= 0 \\ \Rightarrow x - y + 2 = 0 \Rightarrow d_{(0,0)} &= \sqrt{2} < \frac{11}{\sqrt{34}} \\ \text{Also } |z + 5| > |z - 9| \Rightarrow x > 2 &\text{ and} \\ r &= \frac{11}{\sqrt{34}} < 2 \\ |z - 5i| < |z + 9i| \Rightarrow y > -2 \end{aligned}$$

5. Ans (B)

$$\begin{aligned} \frac{x - 1}{1} &= \frac{y - 5}{4} = -\frac{(18)}{17} \\ x &= 1 - \frac{18}{17} = -\frac{1}{17} \\ y &= 5 - \frac{72}{17} = \frac{13}{17} \\ ae &= -\frac{1}{17} \Rightarrow a^2 - b^2 = \frac{1}{289} \\ b^2 &= \frac{13}{17} \\ \Rightarrow a^2 &= \frac{13}{17} + \frac{1}{289} = \frac{222}{289} \\ \Rightarrow a^2 e^2 &= \frac{1}{289} \Rightarrow \frac{222}{289} e^2 = \frac{1}{289} \Rightarrow e^2 = \frac{1}{222} \end{aligned}$$

6. Ans (C)

$$\begin{aligned} \det(A) &= a^2 - bc \leq 0 \\ a^2 &\leq bc \\ \text{If } a = 1 &\Rightarrow bc \geq 1 \Rightarrow {}^5C_2 \times 2! \\ \text{If } a = 2 &\Rightarrow bc \geq 4 \geq 6 + {}^4C_2 \times 2! \\ \text{If } a = 3 &\Rightarrow bc \geq 9 \geq 4 + {}^3C_2 \times 2! \\ \text{If } a = 4 &\Rightarrow bc \geq 16 \Rightarrow 2 + 2 \\ \Rightarrow \text{possible ways} &= 6 + [{}^2C_2 + {}^3C_2 + {}^4C_2 + {}^5C_2] \\ \times 2 &= 6 + 40 = 52 \\ \text{Total ways} &= {}^6C_3 \times {}^3C_1 \times 2 = 20 \times 3 \times 2 = 120 \\ \Rightarrow \text{Probability} &= \frac{52}{120} \end{aligned}$$

7. Ans (B)

$$\begin{aligned} 4k^2 + 4(k - 5)(k + 1) &\leq 0 \\ k^2 + k^2 - 4k - 5 &\leq 0 \\ 2k^2 - 4k - 5 &\leq 0 \\ k &= \frac{4 \pm \sqrt{16 + 40}}{4} = \frac{4 \pm 2\sqrt{14}}{4} = \frac{2 \pm \sqrt{14}}{2} \\ k \in \left[\frac{2 - \sqrt{14}}{2}, \frac{2 + \sqrt{14}}{2} \right] &\Rightarrow m = 3 \\ I &= \int_{-3}^3 [x^3] dx \\ I &= \int_0^3 ([x^3] + [-x^3]) dx = -3 \end{aligned}$$

8. **Ans (D)**

Total such complex numbers are the points inside the circle

$$x^2 + y^2 \leq 100 \text{ with both coordinates integers (b} \neq 0)$$

$$\text{Total} = 10 \times 2 + (9 \times 4 + 8 \times 2 + 7 + 6 + 4) \times 4$$

$$= 20 + 69 \times 4 = 296$$

Possible cases arise when $a = 0$

or

for pythagorean triplets, hence number of possible cases are $16 + 20 = 36$

$$\text{Probability} = \frac{36}{296}$$

9. **Ans (B)**

$$\frac{dy}{dx} - \frac{2xy}{x^2 + 1} = \frac{x^3}{x^2 + 1}$$

$$\text{I.F} = e^{-\int \frac{2x}{x^2+1} dx} = e^{-\ln(x^2+1)} = \frac{1}{x^2+1}$$

$$\Rightarrow y \cdot \frac{1}{x^2+1} = \int \frac{x^3}{(x^2+1)^2} dx$$

$$x^2 + 1 = t$$

$$2x dx = dt$$

$$= \frac{1}{2} \int \frac{(t-1)}{t^2} dt = \frac{1}{2} \int \left(\frac{1}{t} - \frac{1}{t^2} \right) dt$$

$$\Rightarrow \frac{y}{x^2+1} = \frac{1}{2} \left[\ln t + \frac{1}{t} \right] + c$$

$$\Rightarrow y = \frac{1}{2}(x^2+1) \left[\ln(x^2+1) + \frac{1}{x^2+1} \right] + c$$

$$\Rightarrow y = \frac{1}{2} [(x^2+1) \ln(x^2+1) + 1]$$

10. **Ans (A)**

Given statement is fallacy

$$p \Rightarrow q = (\sim p \vee q)$$

$$p \wedge \sim q = \sim (\sim p \vee q)$$

$(\sim p \vee q) \wedge \sim (\sim p \vee q)$ is also fallacy

11. **Ans (B)**

$$\frac{dy}{dx} = x^x (1 + \log x)$$

$$\frac{d^2y}{dx^2} = x^x (1 + \log x)^2 + x^{x-1} = x^x \left((1 + \log x)^2 + \frac{1}{x} \right)$$

$$\frac{d^2x}{dy^2} = \frac{-d^2y/dx^2}{(dy/dx)^3} = - \frac{x^x \left((1 + \log x)^2 + \frac{1}{x} \right)}{x^{3x} (1 + \log x)^3}$$

$$\left(\frac{d^2x}{dy^2} \right)_{x=e} = \frac{-e^e (4 + e^{-1})}{8e^{3e}} = \frac{-(1 + 4e)}{8e^{2e+1}}$$

12. **Ans (C)**

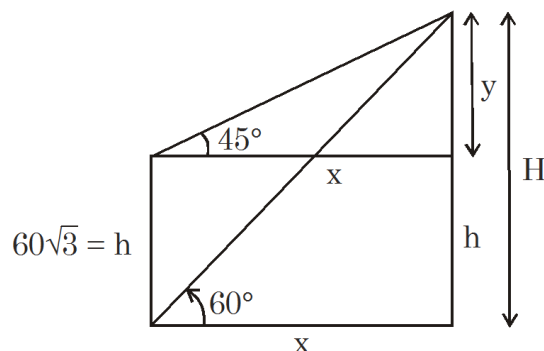
$$\left| e^{nC_0 + nC_1\omega + nC_2\omega^2 + \dots + nC_n\omega^n} \right| = \left| e^{(1+\omega)^n} \right| = \left| e^{(-1)^n\omega^{2n}} \right| = \left| e^{\cos \frac{n\pi}{3} + i \sin \frac{n\pi}{3}} \right|$$

$$= \left| e^{\cos \frac{n\pi}{3}} \right| \text{ can have values}$$

$$= \{e^1, e^{1/2}, e^{-1/2}, e^{-1}\}$$

Four values

13. **Ans (C)**



$$y = x \tan 45^\circ = x$$

$$H = x \tan 60^\circ = \sqrt{3}x$$

$$h = H - y$$

$$60\sqrt{3} = (\sqrt{3} - 1)x$$

$$x = 30\sqrt{3}(\sqrt{3} + 1)$$

$$H = \sqrt{3}x = 90(\sqrt{3} + 1)$$

14. **Ans (C)**

$$(\sin^{-1} x - 2)(\sin^{-1} x - 3) \geq 0 \quad \forall x$$

$$\Rightarrow \sin^{-1} x \geq 1 \Rightarrow x \in [\sin 1, 1]$$

15. **Ans (D)**

$$A = \int_0^1 (x^2 - x^3) dx + \int_1^2 (x^3 - x^2) dx$$

$$= \left[\frac{x^3}{3} - \frac{x^4}{4} \right]_0^1 + \left[\frac{x^4}{4} - \frac{x^3}{3} \right]_1^2$$

$$= \frac{1}{3} - \frac{1}{4} + \left(4 - \frac{8}{3} \right) - \left(\frac{1}{4} - \frac{1}{3} \right)$$

$$= \frac{2}{3} - \frac{1}{2} + 4 - \frac{8}{3} = \frac{7}{2} - 2 = \frac{3}{2}$$

16. **Ans (C)**

For $x > 0$, $f(x) = 2x + \sin x$

$$f'(x) = 2 + \cos x \Rightarrow f'(x) > 0 \text{ and } f \in [0, \infty)$$

For $x < 0$ $f(x) = \sin x$ and $f \in [-1, 0)$

17. **Ans (A)**

Here, $\Delta \neq 0$

$$\text{As, } \Delta = (23 - \alpha) \neq 0$$

18. **Ans (A)**

$$\text{Let } \vec{P} = \lambda(\hat{i} + 2\hat{j} + 3\hat{k}) \times (-2\hat{i} + 4\hat{j} + \hat{k})$$

$$= -10\lambda\hat{i} - 7\lambda\hat{j} + 8\lambda\hat{k}$$

$$\text{given } \vec{P} \cdot (\hat{i} - 2\hat{j} + \hat{k}) = -6$$

$$\Rightarrow -10\lambda + 14\lambda + 8\lambda = -6$$

$$\Rightarrow \lambda = \frac{-1}{2}$$

$$\therefore \vec{P} = 5\hat{i} + \frac{7}{2}\hat{j} - 4\hat{k}$$

$$\Rightarrow \vec{P} \cdot \hat{i} + \vec{P} \cdot \hat{k} = 5 - 4 = 1$$

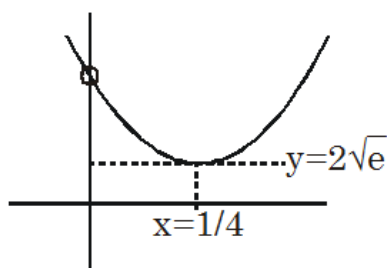
19. **Ans (B)**

$$\lim_{x \rightarrow 0} \frac{\sin 2x + ax^3 + bx}{x^3}$$

$$\lim_{x \rightarrow 0} \frac{2x - \frac{(2x)^3}{3!} + ax^3 + bx}{x^3} = 0$$

$$p + 2 = 0 \quad a - \frac{8}{6} = 0 \quad a = \frac{4}{3}$$

20. **Ans (B)**



$$\frac{e^{2x}}{\sqrt{x}} = K;$$

$$\frac{dy}{dx} = \frac{(4x - 1)e^{2x}}{2x\sqrt{x}}$$

$$\Rightarrow \text{minima at } x = \frac{1}{4} \text{ \&}$$

$$\text{minimum value} = 2\sqrt{e}$$

$$\text{For one solution } K = 2\sqrt{e}$$

PART-3 : MATHEMATICS

SECTION-II

1. **Ans (5)**

$$\frac{(1 - \tan^2 x) dx}{(\sin^4 x (2 - \cos^2 x) + \cos^4 x (2 - \sin^2 x) \sec^2 x)}$$

$$= \int \frac{\cos 2x dx}{2 - 5\sin^2 x \cos^2 x} = \int \frac{4 \cos 2x dx}{8 - 5(\sin 2x)^2}$$

$$\sin 2x = t \Rightarrow 2 \cos 2x dx = dt$$

$$\Rightarrow \int \frac{2dt}{8 - 5t^2} = \frac{2}{5} \int \frac{dt}{\frac{8}{5} - t^2}$$

$$= \frac{2}{5} \times \frac{\sqrt{5}}{2\sqrt{8}} \ln \left(\frac{\frac{\sqrt{8}}{\sqrt{5}} + \sin 2x}{\frac{\sqrt{8}}{\sqrt{5}} - \sin 2x} \right)$$

$$= \frac{1}{2\sqrt{10}} \ln \left(\frac{\sqrt{8} + \sqrt{5} \sin 2x}{\sqrt{8} - \sqrt{5} \sin 2x} \right)$$

2. **Ans (5)**

Coefficient of x^2 in the expansion of

$$(1+x)^2 + (1+x)^3 + \dots + (1+x)^{49} + (1+mx)^{50} \text{ is}$$

$${}^2C_2 + {}^3C_2 + \dots + {}^{49}C_2 + {}^{50}C_2 m^2 = (3n+1) {}^{51}C_3$$

$${}^3C_3 + {}^3C_2 + \dots + {}^{49}C_2 + {}^{50}C_2 m^2 = (3n+1) {}^{51}C_3$$

$${}^{50}C_3 + {}^{50}C_2 m^2 = (3n+1) {}^{51}C_3$$

$$\frac{50.49.48}{6} + \frac{50.49}{2} m^2 = (3n+1) \frac{51.50.49}{6}$$

$$m^2 = 51n + 1$$

$$\text{must be a perfect squared} \Rightarrow n = 5 \text{ and } m = 16$$

$$\text{Ans. } \Rightarrow 5$$

3. **Ans (8)**

$$(\sqrt{7})^{\frac{1}{\log_4 7}} = 7^{\frac{1}{2} \log_7 4} = 4^{\frac{1}{2}} = 2$$

$$\left((\log_2 9)^2 \right)^{\frac{1}{\log_2 (\log_2 9)}}$$

$$\log_2 9 = t, \frac{2}{t \log_2 t} = t^{2 \log_2 t} = 4$$

$$4 \times 2 = 8$$

4. **Ans (4)**

Non-differentiable at $x = 1, 2, 3, 4$ continuous
and differentiable at $x = 5$

5. **Ans (19)**

$$d = 1, a + b + c = 12 \Rightarrow {}^{11}C_2$$

$$d = 2, a + b + c = 19 \Rightarrow {}^8C_2$$

$$d = 3, a + b + c = 6 \Rightarrow {}^5C_2$$

$$\Rightarrow k = 1 + 10 + 28 + 55 = 94 = k$$

$$d = 0 \Rightarrow a + b + c = 15 \Rightarrow {}^{17}C_2$$

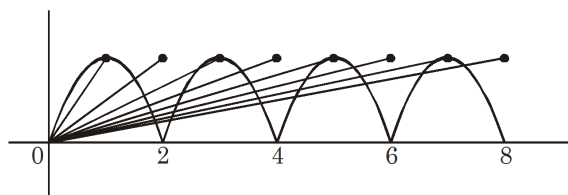
$$d = 1 \Rightarrow a + b + c = 12 \Rightarrow {}^{14}C_2$$

$$d = 2 \Rightarrow a + b + c = 9 \Rightarrow {}^{11}C_2$$

$$d = 3 \Rightarrow a + b + c = 6 \Rightarrow {}^8C_2$$

$$\Rightarrow m - k = {}^{17}C_2 + {}^{14}C_2 = 17 \times 8 + 7 \times 13 \\ = 136 + 91 = 227$$

6. **Ans (32)**



$$\text{Total solutions} = 2(1 + 3 + 5 + 7) = 32$$

7. **Ans (2)**

$$\tan 200^\circ (2 \cot 20^\circ) = 2 \tan 20^\circ \cot 20^\circ = 2$$

8. **Ans (303)**

$$S_K = \frac{K+1}{2}$$

$$\Sigma S_k^2 = \frac{5}{12} A$$

$$\sum_{K=1}^{10} \left(\frac{K+1}{2} \right)^2 = \frac{2^2 + 3^2 + \dots + 11^2}{4} = \frac{5}{12} A$$

$$\frac{11 \times 12 \times 23}{6} - 1 = \frac{5}{3} A$$

$$505 = \frac{5}{3} A, A = 303$$

9. **Ans (1)**

$$f'(x) = 5x^4 + 9x^2 + \frac{1}{1+x^2} + e^x > 0 \quad \forall x \in \mathbb{R}$$

and $f \in (-\infty, \infty) \Rightarrow$ one real root

10. **Ans (2)**

$$-\frac{3\pi}{4} \leq \sin^{-1} x + \tan^{-1} x \leq \frac{3\pi}{4}$$

$$-\frac{3\pi}{4} \leq 2p + 1 \leq \frac{3\pi}{4}$$

$$\frac{-3\pi - 4}{8} \leq p \leq \frac{3\pi - 4}{8}$$

$$-1.68 < p < 0.67$$

$$p = -1, 0$$